Electronic Components Documentation

Project: [Your Project Name]

June 11, 2025

Contents

1 Overview

This document describes the electronic components used in our project, along with their specifications and roles. The system is based on an ESP32 controller and includes servos, PWM drivers, sensors, and switches.

2 Components

2.1 MG996R Servo Motors (x18)

- Type: Digital high torque servo
- Operating Voltage: 4.8V–7.2V
- Stall Torque: Up to 11 kg.cm @ 6V
- Control Signal: PWM (Pulse Width Modulation)
- Application: Movement control (e.g., limbs, joints)

2.2 PCA9685 PWM Controller (x2)

- Channels: 16-channel 12-bit PWM
- Communication: I^2C interface
- Power Supply: 5V logic level
- Used for controlling up to 32 servos with only two I^2C addresses

2.3 POS-150-5-C Power Supply

- Output Voltage: 5V
- Output Current: Up to 30A (150W total)
- Usage: Powers servos and PCA9685 boards

2.4 Micro Endstop Switches (x6)

- Type: Momentary mechanical switches
- Usage: Limit/end position detection
- Triggered when component reaches physical boundary

2.5 ESP32-WROVER-DEV

- Microcontroller with Wi-Fi and Bluetooth
- Extra RAM (4MB PSRAM)
- GPIO, I^2C, SPI support
- Central controller for the system

2.6 MPU6050 Accelerometers (x2)

- 3-axis gyroscope and 3-axis accelerometer
- Communication: I^2C
- Used for motion sensing and orientation

3 Electrical Wiring Diagram

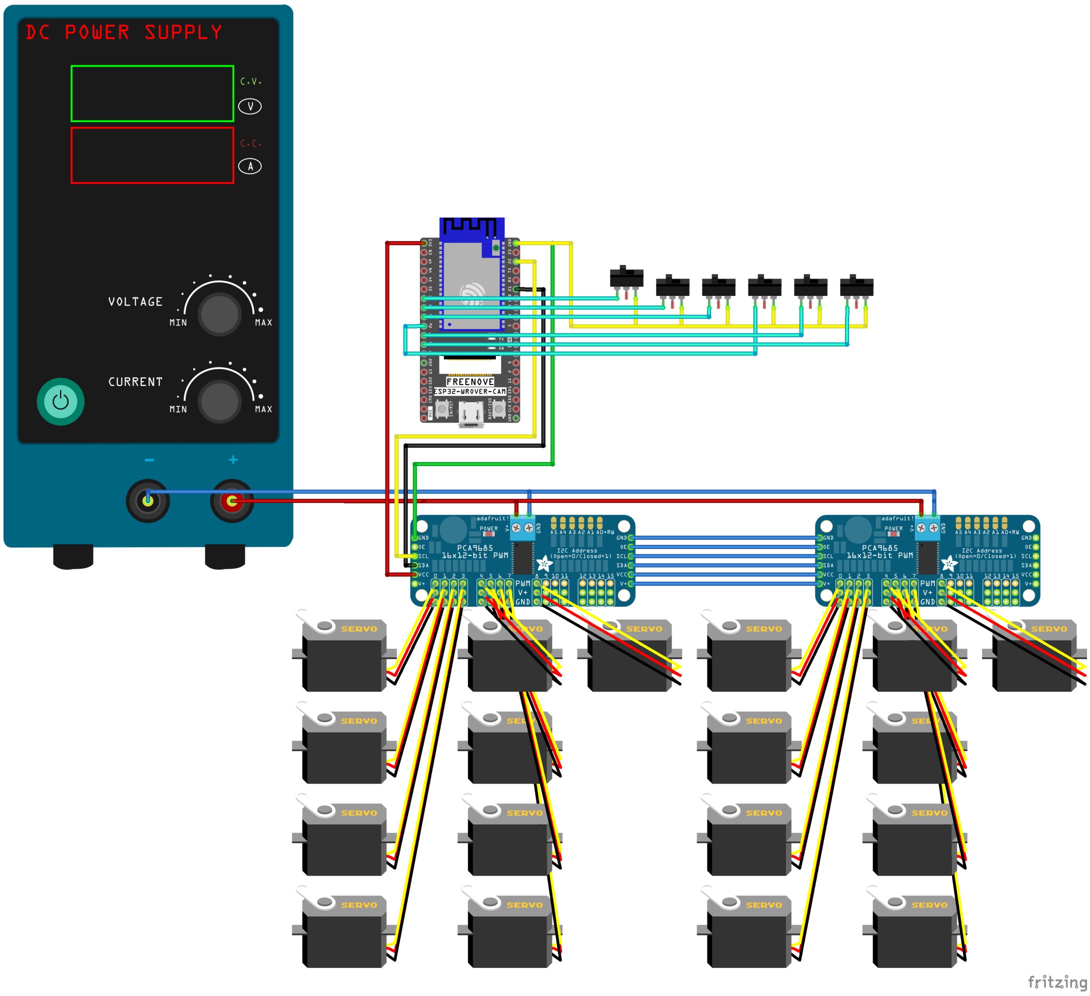

Figure 1: Wiring Diagram of the Electronic System